VINAY SANTOSH S.

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MECHANICAL ENGINEERING & MBA | LEAN SIX SIGMA GREEN BELT

Goal-oriented professional bridging mechanical execution and business strategy. Proven expertise in CAD design, Finite Element Analysis (FEA), and industrial fabrication, complemented by an MBA focus on operations management and supply chain optimization. Dedicated to leveraging data analytics, lean principles, and robust engineering methodologies to drive operational efficiency in EPC and manufacturing environments.

EDUCATION

Master of Business Administration (MBA) — Operations Management *Expected 2027*
St. Joseph's Institute of Technology

Bachelor of Engineering in Mechanical Engineering *2024*
St. Joseph's College of Engineering — 8.15 CGPA

INDUSTRIAL TRAINING & APPLIED EXPERIENCE

Mechanical Engineering In-Plant Trainee (FEA) *In-Plant Training*
Indira Gandhi Centre for Atomic Research (IGCAR) — Kalpakkam, TN

- Completed technical training program at a premier government research institute focused on Finite Element Method (FEM).
- Conducted rigorous structural analysis on mechanical components utilizing advanced Abaqus simulation software.
- Evaluated component integrity and performance, establishing a strong practical foundation in advanced FEA techniques.

CORE PROJECTS & RESEARCH

Parametric Design Optimization of Road Car Aerodynamics *Academic Research*

- Co-authored a CFD study investigating the correlation between wing gap variance, vortex generators, and aerodynamic stability.
- Utilized ANSYS Workbench Genetic Aggregation algorithms to evaluate Angle of Attack (AoA) and reduce drag coefficients.

Supply Chain Optimization & Logistics Analytics *Data Analytics Project*

- Analyzed 180,000+ shipment records using the DataCo dataset to identify root causes of regional delivery delays.
- Applied Lean Six Sigma principles to design a Power BI framework projecting a 15% reduction in late delivery risks.

Strategic Operations & Financial Analysis: TNPL *MBA Case Study*

- Conducted a comprehensive operational and financial analysis of TNPL, evaluating its lean-green manufacturing synergy.
- Analyzed capacity utilization, DuPont financial metrics, and the circular-economy paper production process using bagasse.

Automated Pneumatic Sheet Metal Cutter *Fabrication Project*

- Designed and fabricated a compact, automated sheet metal cutter actuated by a single pneumatic cylinder mechanism.
- Improved operator safety metrics and dimensional cutting accuracy for thin-sheet precision compared to hydraulic models.

SAE Drone Development & Aerodynamic Design *Engineering Competition*

- Served as Wing Design Specialist, utilizing SolidWorks for the 3D modeling of complex aerodynamic components.
- Secured 8th Place in the Southern Region following rigorous operational flight testing and performance evaluations.

LEADERSHIP & SYMPOSIA COORDINATION

Event Coordinator — Epicstra 2025 Symposium *St. Joseph's Institute of Technology*

- Spearheaded the Operations Management track, facilitating interactive simulations and competitive activities.
- Evaluated participating teams based on strategic thinking, operational efficiency, and process optimization execution.

Event Coordinator — JetMech Symposium *St. Joseph's College of Engineering*

- Managed technical tracks, coordinated logistics, and led cross-functional student teams to execute the department symposium.

HONORS, CERTIFICATIONS & ACTIVITIES

- **Honors:** State Qualifier, TN Skills Additive Manufacturing Competition. "Best Manager" competitor, Daksha 25 Symposium.
- **Certifications:** Lean Six Sigma Green Belt (MSME) | Financial Markets (NISM Series V-A) | CATIA V5 | 4D Printing.
- **Advanced Workshops:** Industrial Robotics & Automation | Data Analytics & Power BI | VR Simulation (Unity).

TECHNICAL SKILLS

ENGINEERING & SIMULATION	OPERATIONS & STRATEGY	DATA & SOFTWARE
SolidWorks & CATIA V5	Operations Management	Power BI
AutoCAD & 3D Modeling	Lean Six Sigma	Data Analytics
ANSYS Workbench	Theory of Constraints	Microsoft Excel
Abaqus (FEA)	Supply Chain Optimization	Unity Engine (VR Simulation)
Pneumatics & Fabrication	Process Improvement	Cross-Functional Leadership

LANGUAGES

English (Fluent) • **Tamil** (Native) • **Hindi** (Conversational)